

Problem

Fix

Goal

Overfitting
memorises the
training data

Early stopping
Note 1022

Dropout
Note 1024–1025

L1 and L2
regularisation
Note 1026

Unstable gradients
vanishing or exploding
Note 1018

Better activations
Note 1027

ReLU and
its variants
Note 1028

Weight initialisation
Note 1029

Xavier and He
Note 1030

Batch normalisation
Note 1031

Backpropagation
Note 1015–1017

Slow training
poor or wobbly
convergence

Scale the inputs
Note 1023

Choose the batch size
Note 1020

Better optimizers
see the optimizers map

**A better
neural network**
Note 1021

→ fixes

→ is a kind of / leads to

→ makes the network better

from another map